



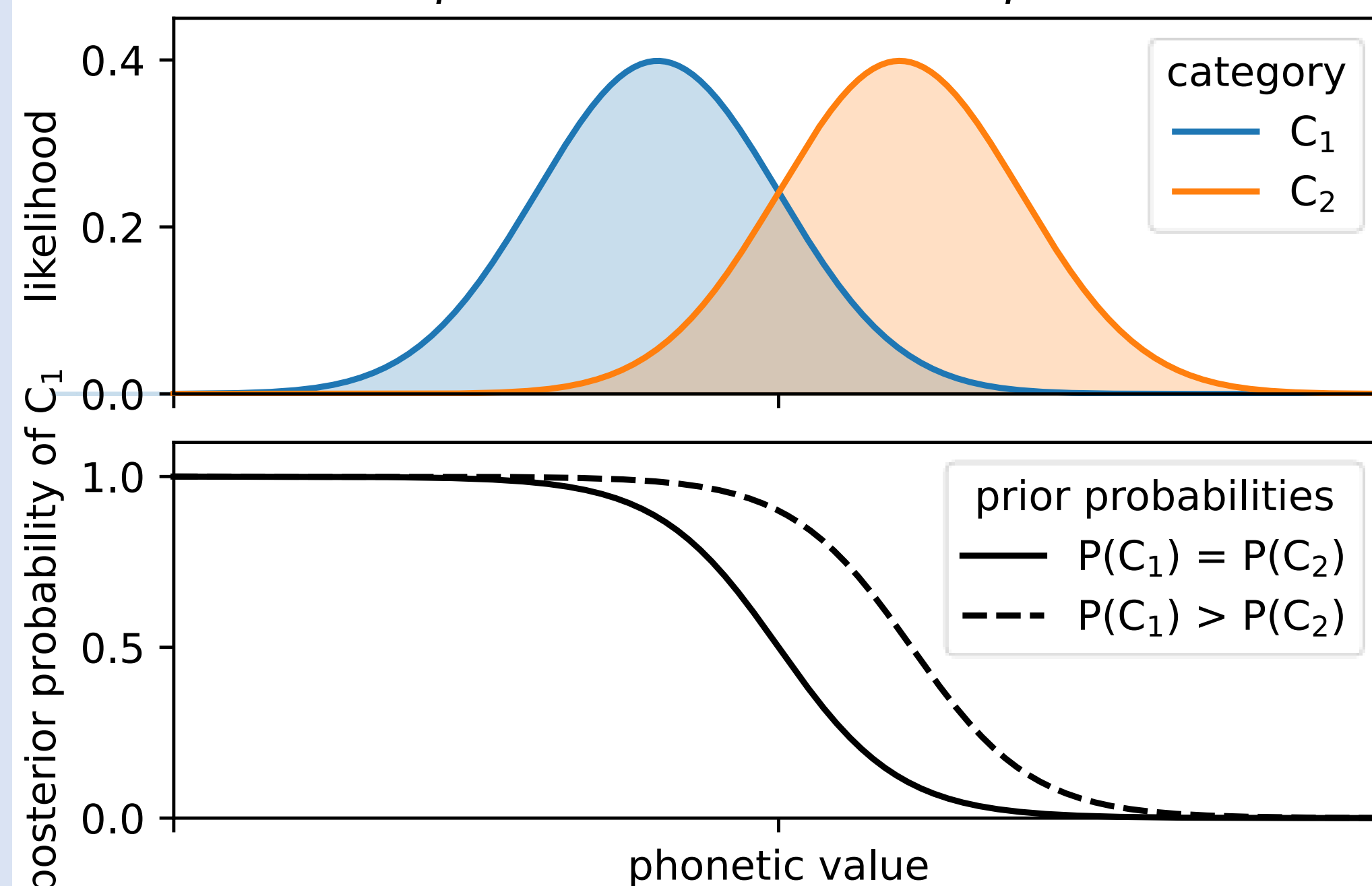
2pSC28: A Bayesian Account of Perceptual Asymmetries in Confusion Matrix Data

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Bayesian Speech Perception

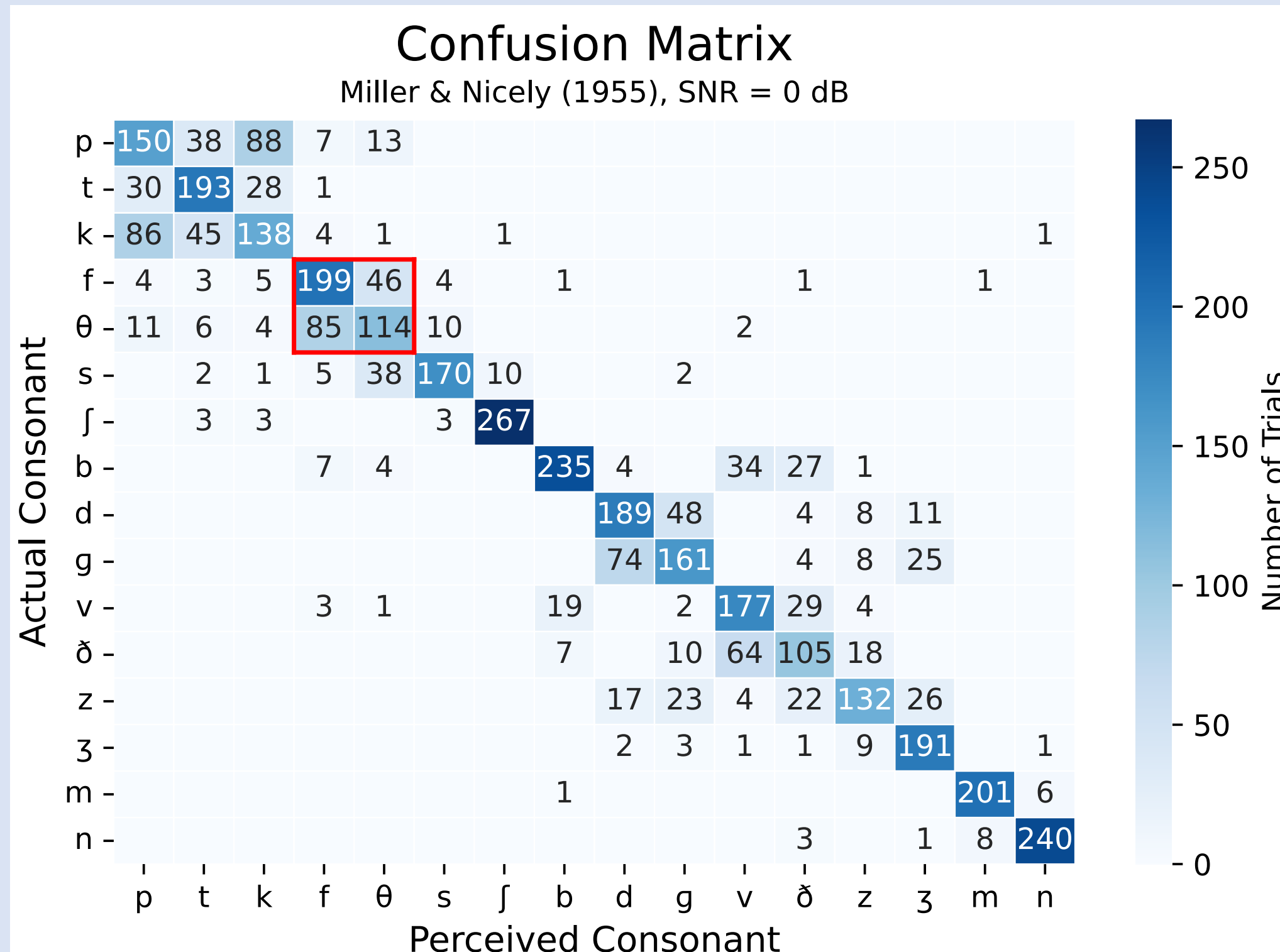
Human speech perception appears well-captured by Bayesian models of optimal listeners
(Norris & McQueen 2008, Clayards et al. 2008; Feldman et al., 2009)

Phonetic categorization tracks posterior probability
 $\text{posterior} \propto \text{likelihood} \times \text{prior}$



Prediction: prior bias toward more frequent categories

Phoneme Confusion Matrices



Consonant categorization within CVs over many trials

Confusion between Cs induced by adding white noise

- signal-to-noise ratio (SNR):**
SNR = 0 dB → CV signal and noise are equally loud

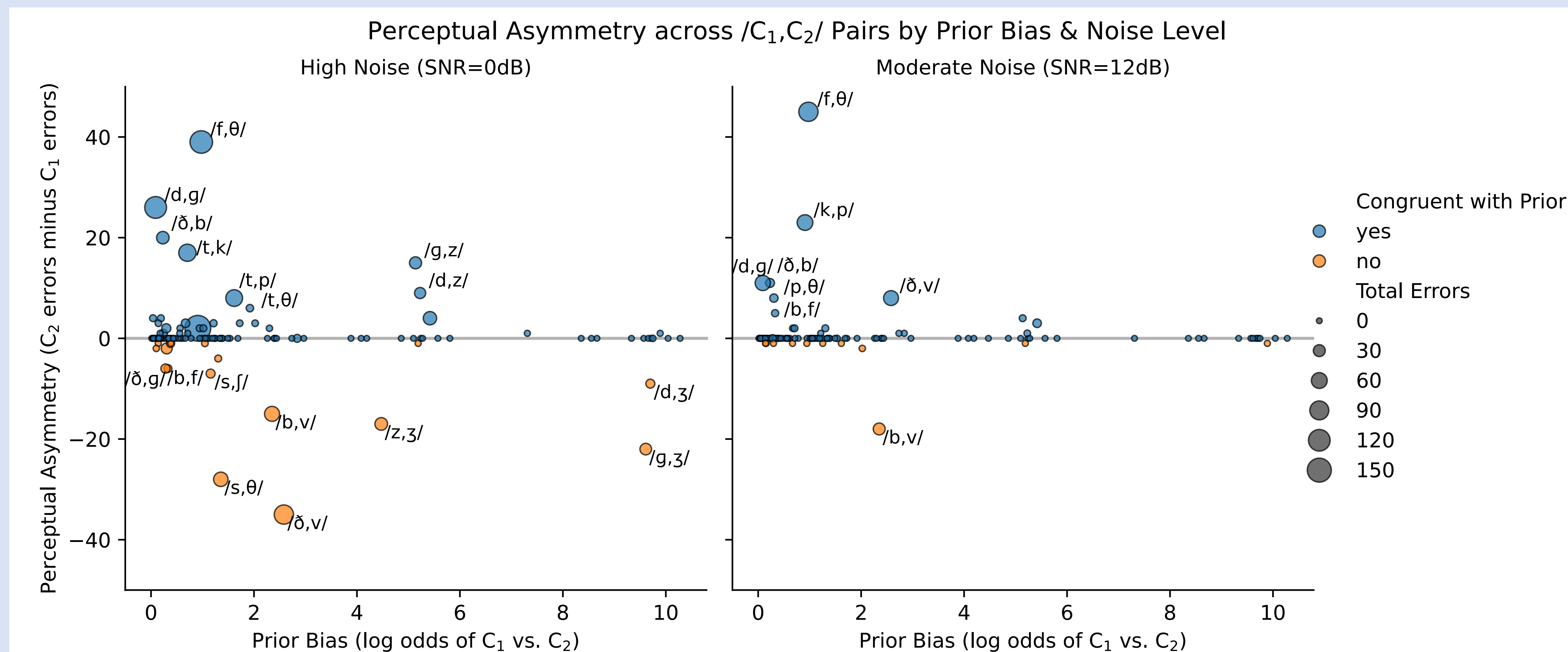
Perceptual asymmetries:

- e.g. more /θ/ misperceived as /f/ than vice versa, consistent with prior bias since $P(/f/) > P(/θ/)$

Role of Prior Bias

Approach: check perceptual asymmetries against prior bias using Miller & Nicely's (1955) confusion matrices

- Prior bias estimated from token frequency of word-initial Cs before stressed vowel
 - SublexUS lexical frequencies (Brysbaert & New, 2009) and CMUdict pronunciations (Weide, 1998)



High noise

- 65% (22/34) asymmetries larger than 1 match prior bias

Moderate noise

- 86% (12/14) asymmetries larger than 1 match prior bias

Takeaway: prior bias can only partially account for perceptual asymmetries

Role of Likelihood Distortion

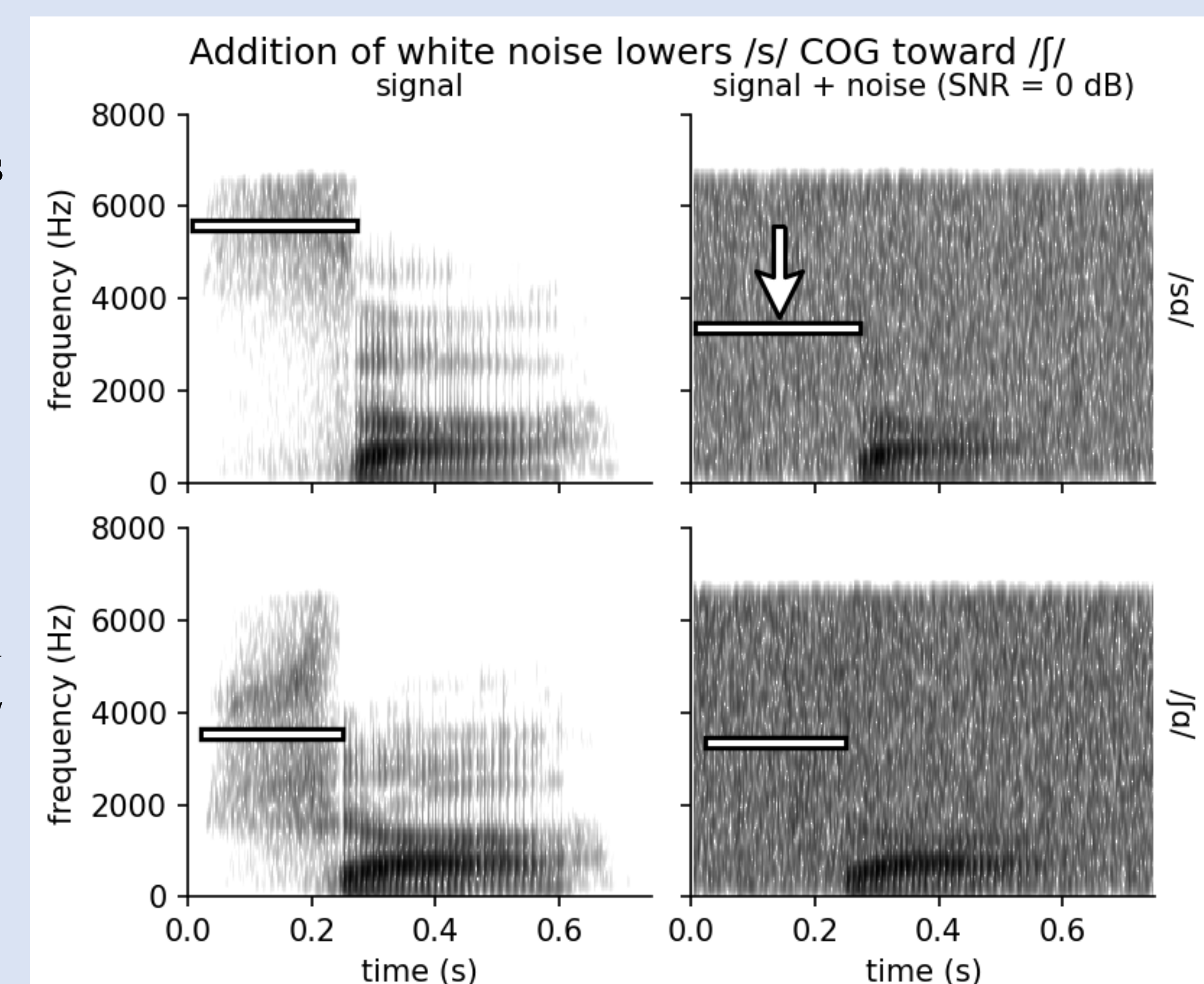
Insight 1: lower noise → more consistent with prior bias

- Insight 2:** most prior-incongruent asymmetries favor fricatives
- fricative signal is noise

- Explanation:** stimulus noise distorts acoustic likelihoods
- toward diffuse fricatives with low center of gravity (COG)

- Example:** high noise yields prior-incongruent /s, f/ asymmetry
- explained because noise lowers COG of /s/ signal toward /f/

Takeaway: confusion reflects perceptual similarity not only between phonemes, but also between phonemes and noise



Conclusions

Confusion matrices provide rich perceptual data worth detailed analysis

Asymmetries are consistent with Bayesian perception:

- most are congruent with prior bias
- others explained by likelihood distortion from noise

Acoustic distortion raises doubt about validity of perceptual similarity measures beyond noisy contexts

Future Work

Analysis of confusion matrices with other noise types

- e.g. Cutler et al. (2004) use multi-speaker babble

Explicit modeling of acoustic likelihood distributions

- likelihoods from HMM/GMM acoustic models
- simulating confusion matrix data with generative statistical models to assess role of priors/likelihoods
- perceptual similarity measures from acoustic models, including for low-resource languages

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